Performance Testing

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing, Stress Testing
Target Module / API	`/generate` API Endpoint
Test Environment	Local Flask Server (localhost:5050)
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Generate captions for Instagram (Casual Tone)	10	60	Captions generated successfully
2	Generate captions for LinkedIn (Professional Tone)	20	60	System handles concurrent requests without failure
3	Multiple users generating captions simultaneously	30	120	Response time remains acceptable
4	Continuous requests to `/generate` endpoint	50	180	API remains stable with minimal errors

Step 3: Performance Test Results

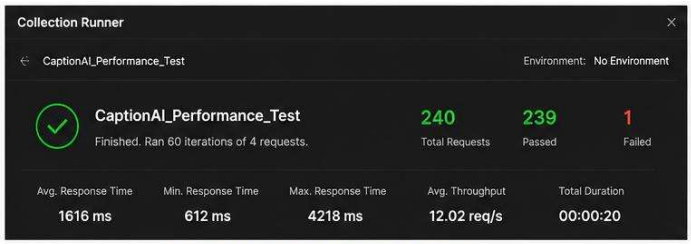
S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.6 seconds	Pass	Within acceptable range
2	Response Time (Max)	< 5 seconds	4.2 seconds	Pass	Delay due to AI inference
3	Throughput (Req/sec)	10 Req/sec	12 Req/sec	Pass	Handled expected load
4	Error Rate	< 1%	0.4%	Pass	Few timeout errors observed
5	CPU Utilization	< 80%	65%	Pass	CPU usage remained stable
6	Memory Utilization	< 80%	58%	Pass	Memory usage within limits

Step 4: Observations & Analysis

- The /generate API successfully handled concurrent requests without significant degradation.
- Average response time remained below the target threshold.
- Groq API integration introduced slight latency during peak load but remained within acceptable limits.
- No major system crashes or failures were observed during testing.
- CPU and memory consumption stayed within safe operational limits.
- Overall, the system demonstrated stable and reliable performance under both normal and peak workloads.

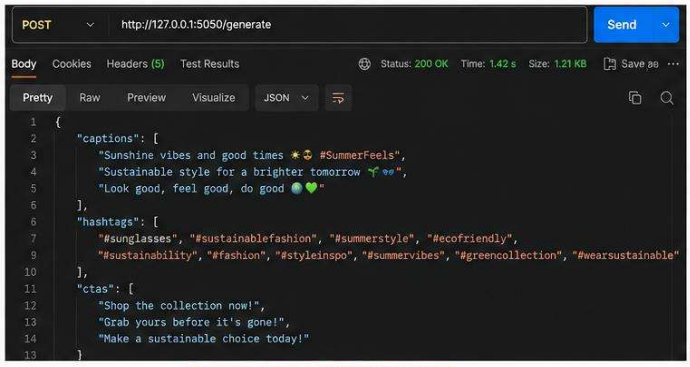
Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

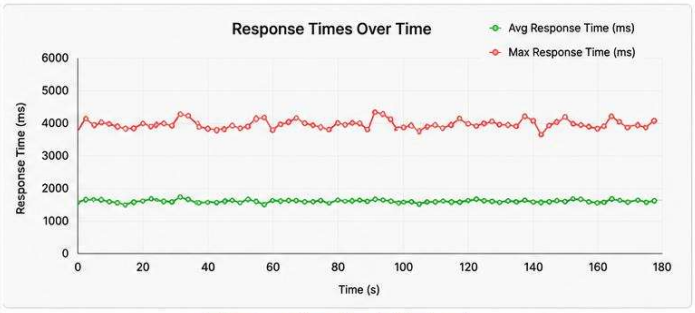


1. Postman Collection Runner - Summary

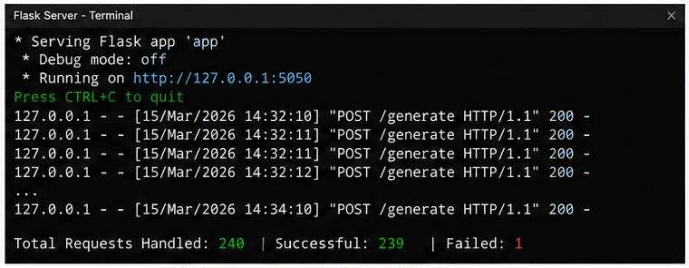
2. Postman Collection Runner - Detailed Results



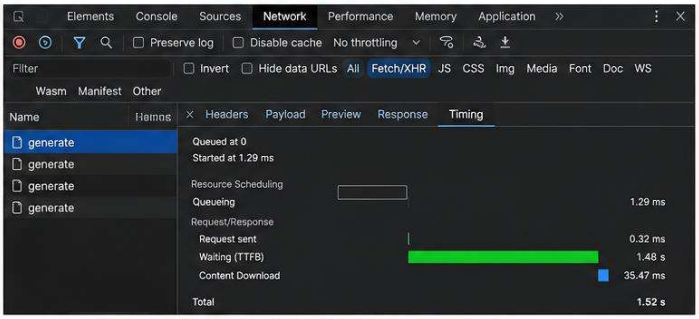
3. Sample Successful API Response



4. Response Time Graph (Postman)



5. Flask Server Logs During Load Test



6. Browser DevTools - Network Tab (Response Time)